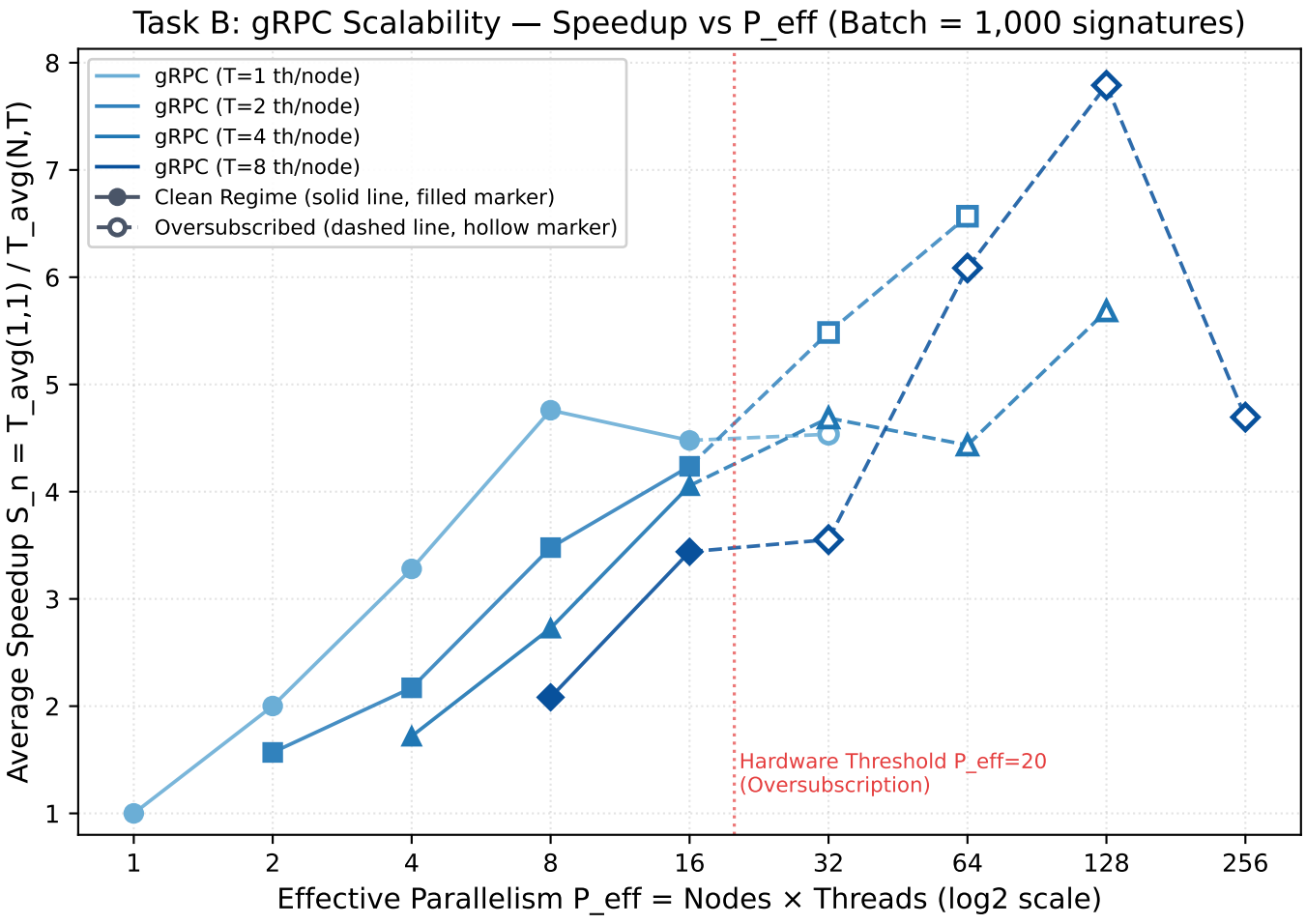
Task B: gRPC Scalability — Speedup vs P_eff (Batch = 1,000 signatures)
gRPC (T=1 th/node)
gRPC (T=2 th/node)
gRPC (T=4 th/node)
gRPC (T=8 th/node)
Clean Regime (solid line, filled marker)
Oversubscribed (dashed line, hollow marker)
Hardware Threshold P_eff=20
(Oversubscription)
Average Speedup S_n = T_avg(1,1) / T_avg(N,T)
Effective Parallelism P_eff = Nodes × Threads (log2 scale)
1
2
3
4
5
6
7
8
1
2
4
8
16
32
64
128
256